TEXORA · LECTURE SHEET

The Simplex Machine (Speed Frame)

The roving frame: it drafts draw-frame sliver down to a fine roving, protects that roving with a small amount of twist, and winds it in a carefully built package on a bobbin — the last preparation before the ring frame, and the machine whose cone drums and differential motion are the classic study in spinning machinery.

TE-2201 · SPINNING TECHNOLOGY-II · TOPIC 3 · LEVEL 2 TERM 2

1. Principles and objects

Draw-frame sliver is still far too coarse to feed to a ring frame: the ring frame's draft (about 10–60) cannot bridge the gap alone without producing an uncontrollably uneven strand, and the sliver has no twist, so it could not be wound, transported or unwound without damage. The simplex — also called the **speed frame** or **roving frame** — closes the gap. Its objects:

- **Attenuation:** to draft the sliver (draft roughly 5–25) down to a **roving** of the hank (count) the ring frame needs.
- **Twisting:** to insert a small, carefully chosen amount of twist so the roving has enough strength to be wound, handled and unwound — but not so much that the ring frame cannot draft it out again.
- **Winding:** to wind the roving onto bobbins in a built package (parallel coils in layers, tapered at the ends) that unwinds cleanly at high speed on the ring frame creel.

2. The drafting system in detail

Roving-frame drafting must control fibres that the draw frame has already made fine and parallel; it therefore uses **apron drafting**. The common arrangements:

SYSTEM	CONSTRUCTION	NOTES
3-over-3 roller, double apron	Three lines of bottom rollers with three top rollers; the middle pair works through a pair of leather/synthetic **aprons** that carry the fibres almost to the front roller nip	The modern standard for cotton and blends: the apron zone controls floating fibres, giving the most even roving at high draft
4-over-4 roller	Four roller pairs, often with aprons at the front zone and a condenser guide between zones	Used for longer staples and higher total drafts; two drafting zones share the work

2-zone and SKF-type high-draft variants	Apron zone plus a break-draft zone, with cradles that set the apron gap precisely	Optimised for man-made fibres and very fine rovings

Amount and distribution of draft. Total draft is split between a small **break draft** (about 1.1–1.4, between back and middle rollers — it merely tensions the sliver and breaks its false cohesion) and the **main draft** (the remainder, in the apron-controlled front zone, where evenness is decided). Roller settings (the gauge between roller nips) are set a few millimetres above the fibre staple length in each zone: too wide and fibres float out of control (drafting wave, thick and thin places); too close and fibres are crushed, cut, or drafted in jerks. Top rollers are loaded by spring, dead weight or pneumatic pressure so the nips grip without crushing.

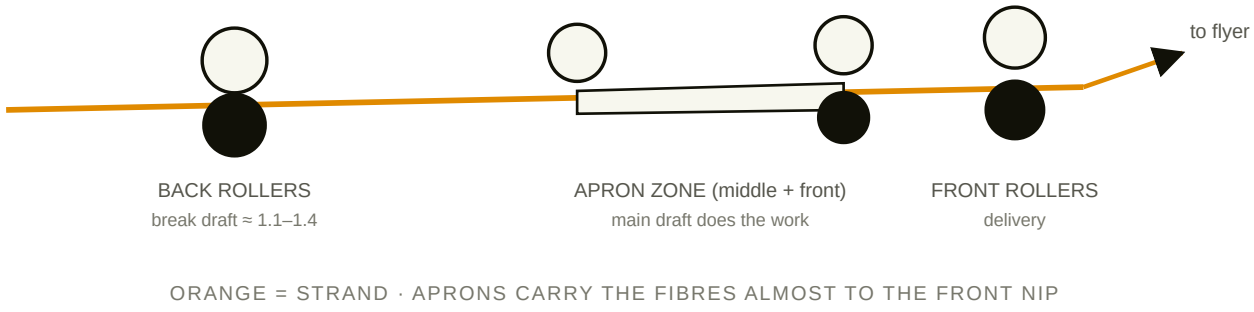


Figure 1 — A 3-over-3 apron drafting system. The break draft tensions the strand; the main draft, inside the apron-controlled zone, decides roving evenness.

3. Twisting — how much, and why it differs by material

Twist is inserted by the rotating **flyer**: each flyer revolution puts one turn into the roving between the front roller and the winding point. Twist is specified by the **twist multiplier (TM)**:

$$\text{Turns per inch (TPI)} = \text{TM} \times \sqrt{(\text{roving hank, Ne})}$$

Roving needs only enough twist to cohere: TM for cotton roving is roughly 0.8–1.4 — fine rovings and longer staples toward the lower values (long fibres cohere better and the ring frame must still draft the roving), coarse rovings toward the higher. **Different materials need different amounts:** smooth, slippery man-made fibres need *more* twist than cotton of the same hank because their fibres grip less; wool's scaly fibres cohere readily and need less; very short-staple cottons need more. Too much roving twist survives into the ring-frame drafting zone and causes hard ends and uneven yarn; too little and the roving stretches or breaks in the creel.

False twist. On its way to the flyer the roving is often passed over a false-twist device or through the flyer top in a path that temporarily twists the strand and then lets the twist run out again. The temporary twist compacts the roving in the vulnerable zone between front roller and flyer — fewer broken ends, less fly — while the *permanent* twist in the wound roving remains only what the flyer legitimately inserted. False twist therefore buys strength exactly where the strand is weakest, without paying for it in ring-frame draftability.

4. Winding and bobbin building

Winding is the machine's subtle problem. The flyer rotates at (nearly) constant speed and the front rollers deliver roving at constant length per minute. The bobbin must take up exactly that length on a package whose diameter grows from about 40 mm (empty) to 110–130 mm (full). Length wound per minute = bobbin surface speed relative to the flyer, so:

Bobbin rpm must fall in inverse proportion to package diameter as the bobbin fills

The classical simplex uses **bobbin lead**: the bobbin rotates slightly faster than the flyer, and the difference — the winding speed — lays the roving on. The bobbin is driven through the **differential motion** (Section 5) so its speed can be varied continuously while the flyer speed stays fixed. As each layer completes, the **builder motion** does three things together: it reverses the traverse of the bobbin rail, it **shortens the traverse slightly** (so the package ends taper into the conical shape that cannot slough off), and it shifts the belt on the cone drums to slow the bobbin drive for the larger diameter.

4.1 The bobbin rail and the spindles

The bobbins sit on **spindles** carried by the lifting **bobbin rail**, which traverses up and down past the flyers, laying successive coils side by side. The rail's stroke defines the package height; its speed must fall as diameter grows (the same length of roving covers fewer turns on a larger bobbin), which is why rail speed is also taken from the variable cone-drum drive. Spindles run in the rail with their own bearings; flyers — mounted on a stationary top rail — are driven at constant speed from the main shaft, each flyer's presser finger guiding the roving onto the bobbin as it rotates.

5. Cone drum drive and the differential motion

Object. Two independent requirements meet at the bobbin: its speed must follow the package diameter, and the machine must keep flyer and rollers at constant speed. One input alone cannot do both.

Cone drums. A pair of opposed conical drums (one driver, one driven, their tapers reversed) carries a belt whose position along the drums sets the speed ratio continuously: belt at the wide end of the driver = fast output; belt shifted toward the narrow end = slow output. The builder motion shifts the belt a

small step at every layer change, so the driven speed falls steadily through the doff — a purely mechanical analogue computer for the diameter/speed relation.

Differential motion. The bobbin drive combines two rotations in an epicyclic (differential) gear: one input from the main shaft at constant speed, and a second input from the cone drums at the variable speed. The gear *adds* them, so the bobbin receives:

$$\text{Bobbin speed} = \text{constant (main) component} + \text{variable (cone drum) component}$$

At the start of a doff the cone-drum component is large (small bobbin needs high rpm); as the belt travels, that component shrinks, and the bobbin slows exactly as the diameter demands — while the flyer never changes speed. Understand this one mechanism and the whole machine's gearing falls into place.

5.1 Alternative methods of bobbin drive

- **Separate electric drives (modern machines):** independent servo or inverter-controlled motors for flyers, rollers and bobbins, with the diameter/speed law computed electronically — no cone drums, no differential, settings changed from a screen, and winding tension held more exactly.
- **Flyer lead:** winding with the flyer faster than the bobbin (used more in wool and some long-staple frames); the package builds under lower tension but needs the flyer speed varied instead.
- **Individual spindle/flyer motor drives** on the newest frames remove long line shafts, letting each unit be stopped and set independently.

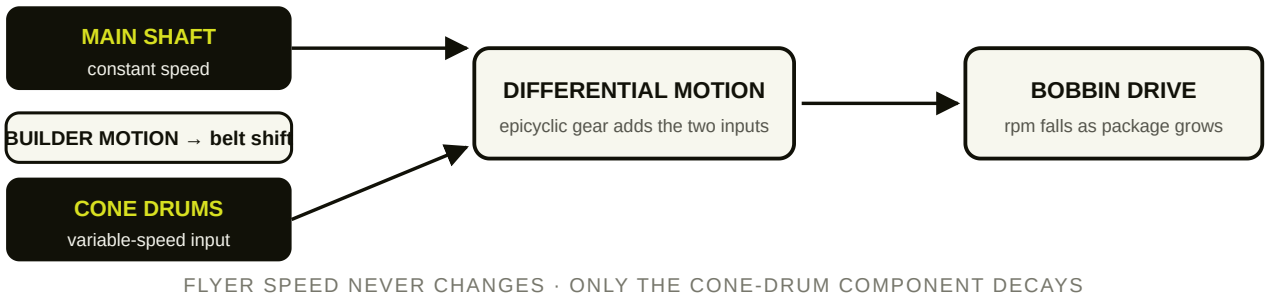


Figure 2 — The classical bobbin drive: the differential adds a constant main-shaft input to a cone-drum input that the builder motion reduces, layer by layer, as the package diameter grows.

6. Settings, speeds and production

Principal settings: roller gauges (from staple length), draft change gears (from sliver hank and roving hank required), twist gear (from TPI required), builder and cone-belt start position (from bobbin size), and rail traverse length. Flyer speeds run about 800–1,500 rpm for cotton roving frames (modern machines higher), limited by winding tension and package stability rather than by the drive.

WORKED EXAMPLE — TWIST AND DELIVERY ON THE SIMPLEX

A roving of 1.5 hank (Ne) is spun with TM = 1.0.

TPI = 1.0 × √1.5 ≈ **1.22 turns/inch**.

With the flyer at 1,000 rpm, front-roller delivery = flyer rpm ÷ TPI = 1000 ÷ 1.22 ≈ **800 in/min ≈ 20.3 m/min** — one turn is inserted per inch delivered.

Per 8-hour shift at 85 % efficiency: 800 × 60 × 8 × 0.85 = 326,400 in ≈ 9,067 yd ≈ 10.8 hanks per spindle. Multiply by the spindle count for frame production.

Glossary

Roving — the drafted, lightly twisted strand delivered by the simplex, specified by its hank (840-yd lengths per pound). **Simplex / speed frame** — the roving frame itself. **Flyer** — the rotating inverted-U component that inserts twist and guides the roving to the bobbin. **Break draft** — the small preliminary draft in the back zone. **Apron drafting** — drafting in which fibre aprons control floating fibres up to the front nip. **Differential motion** — the epicyclic gear that adds a constant and a variable rotation to drive the bobbins. **Cone drums** — opposed conical drums with a travelling belt giving a continuously variable speed ratio. **Builder motion** — the mechanism that reverses and shortens the rail traverse and shifts the cone belt at each layer. **Bobbin lead** — winding arrangement in which the bobbin rotates faster than the flyer.

Key points

• The simplex drafts sliver by 5–25, adds minimal twist (TM ≈ 0.8–1.4), and builds a precision roving package.

• Apron drafting controls floating fibres; the draft splits into a small break draft and the main front-zone draft.

• Roving twist is a compromise: enough to handle, little enough to draft out again; slippery fibres need more, cohesive fibres less. False twist strengthens only the weak delivery zone.

• Winding demands bobbin rpm ∝ 1/package diameter — delivered classically by cone drums plus the differential motion, today often by independent electronic drives.

• The builder motion shortens each traverse, tapering the package ends, and steps the cone belt at every layer.

• TPI = TM × √hank; delivery (in/min) = flyer rpm ÷ TPI.

Practice

1. For a 2.0 hank roving with TM 1.1, calculate the TPI, and the front-roller delivery (m/min) if the flyer runs at 1,200 rpm.
2. A roving package grows from 45 mm to 120 mm diameter during a doff. If the bobbin starts at 1,150 rpm, at what rpm must it finish (flyer speed constant, bobbin lead)? Show the relation used.
3. Explain, in gear terms, why neither the main shaft alone nor the cone drums alone can drive the bobbins correctly.
4. A mill changes from cotton to a smooth polyester sliver of the same hank and finds rovings stretching and breaking in the ring-frame creel. What has gone wrong, and what single change fixes it?
5. Why is the break draft kept near 1.1–1.4 instead of taking half the total draft in the back zone?
6. List the three simultaneous actions of the builder motion at each layer change.

Answer key

1. TPI = 1.1 × √2.0 ≈ 1.1 × 1.414 ≈ **1.56 tpi**. Delivery = 1200 ÷ 1.56 ≈ 771 in/min ≈ **19.6 m/min**.
2. Winding surface speed must stay constant and equal to delivery, so bobbin rpm ∝ 1/diameter: final rpm = 1150 × 45/120 ≈ **431 rpm**.
3. The bobbin needs a speed = a constant base plus a steadily decreasing winding component. The main shaft supplies only constant speed; the cone drums supply a variable speed but with no constant base referenced to flyer rotation. The epicyclic differential adds the two inputs, producing exactly the required law while the flyer runs unchanged.
4. Polyester fibres are smooth and grip poorly, so at cotton's twist multiplier the roving lacks cohesion. Raise the twist (higher TM, via the twist gear) — the roving gains strength for handling and unwinding; check that ring-frame drafting still opens it (draft up slightly if ends go hard).
5. The back zone has no aprons and poor fibre control; putting real draft there would generate drafting wave and thick/thin places that the front zone cannot remove. The break draft only tensions the strand and straightens it; all genuine attenuation belongs in the apron-controlled zone where floating fibres are held.
6. (i) Reverse the bobbin-rail traverse; (ii) shorten the traverse slightly so package ends taper; (iii) shift the belt along the cone drums to reduce the bobbin drive speed for the new, larger diameter.
